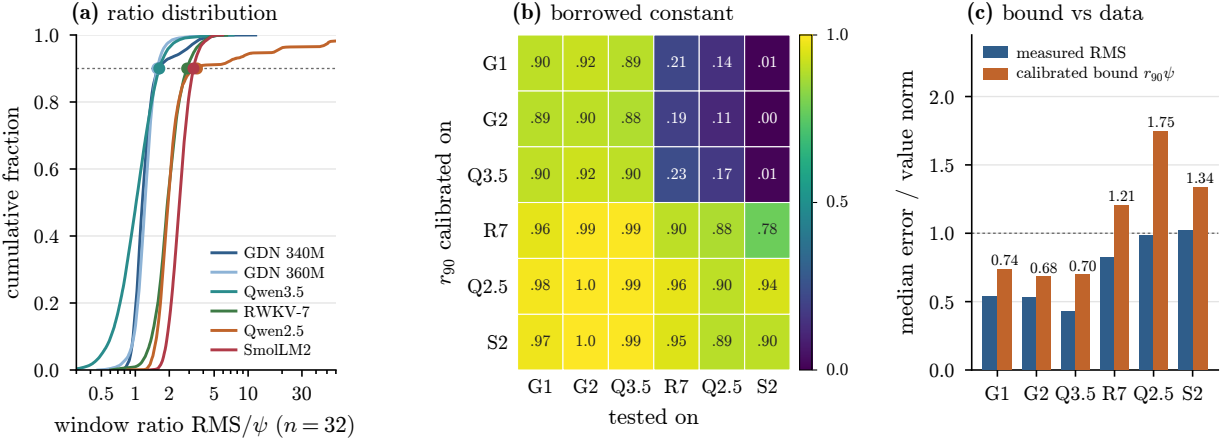
(a) ratio distribution
cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1
2
5
10
30
window ratio RMS/ψ (n = 32)
GDN 340M
GDN 360M
Qwen3.5
RWKV-7
Qwen2.5
SmolLM2
(b) borrowed constant
r90 calibrated on
G1 .90 .92 .89 .21 .14 .01
G2 .89 .90 .88 .19 .11 .00
Q3.5 .90 .92 .90 .23 .17 .01
R7 .96 .99 .99 .90 .88 .78
Q2.5 .98 1.0 .99 .96 .90 .94
S2 .97 1.0 .99 .95 .89 .90
G1 G2 Q3.5 R7 Q2.5 S2
tested on
1.0
0.5
0.0
(c) bound vs data
measured RMS
calibrated bound r90ψ
median error / value norm
2.0
1.5
1.0
0.5
0.0
0.74
0.68
0.70
1.21
1.75
1.34
G1 G2 Q3.5 R7 Q2.5 S2